GridSense AI - handoff report

2026-11-09. Both repos frozen on branch wip/handoff-2026-11-09. main is untouched and still what the deployed API and Vercel site run, so the app being shown to reviewers is unaffected by anything below.

Where the code is

Backend: branch wip/handoff-2026-11-09 at 8f2285b (work commit 3a61618 + HANDOFF.md). main = 3853127 = origin/main, already pushed. 35 files, +5741/-95.

Frontend: branch wip/handoff-2026-11-09 at 11f10fc (work commit b6324f8 + HANDOFF.md). main = 2d630e6 = origin/main. 5 files, +109/-16.

No draft PR could be opened: the gh CLI is not installed here. Open them from github.com/huzaifahimad/Grid_Sense.AI-Backend-/pull/new/wip/handoff-2026-11-09 and github.com/huzaifahimad/Grid_Sense.AI-frontend-/pull/new/wip/handoff-2026-11-09.

Checks actually run (do not read these as 'all green')

- Backend pytest: 157 passed, 6 skipped in 187.83 s. This is a pass.
- scripts/verify_prediction_integrity.py: 27 passed, 6 FAILED of 33. Not a model regression: it compares the live API, whose origin has advanced to 2026-11-09T08:00Z, against local artifacts that end 2026-11-04T17:00Z. Three 'served origin is the true end of gap-free history' and three 'independent recursion reproduces served p50' fail from that one cause. The verifier needs to learn about the live/committed split before it can be a CI gate. I did not edit the check.
- Frontend npm run build: succeeded, 865 modules, 14.62 s. No browser verification of this branch - the defects were reproduced before the fix; the fix is build-checked only.

Done

- Model choice measured: LightGBM stays served; Random Forest's h+48 band covers 0.7458 against 0.80 nominal, which on an alarm product is a manufactured miss.
- Split leak quantified: 80.53% of the honest test window sits inside a shuffled 80% train partition and 99.06% have their own t-1h hour in training, inflating MAPE 9-17%.
- Input smoothing (recipe step 9) measured as a negative and rejected: 4 variants, paired bootstrap, +7.23 pp worse at h+1 for the most aggressive variant.
- Train/serve parity proven; an audit's train/serve-skew 'P0' is refuted with the exact frame the estimator receives (6 features x 4 targets, all differences 0.0).
- predict_live CLI (step 8) runs end to end; live accuracy ledger (step 11) and a gated weekly retrain/promote chain (step 10) exist with tests.
- Risk panel now scores the newest complete row, additively (snapshot_mode, snapshot_age_hours), so the deployed frontend keeps working.
- Weather-lag claims corrected to the measured sawtooth with dated readings: 90 h in the committed frame, 134.0 h at 2026-11-06T15:29Z, 55.0 h at 2026-11-07T07:52Z.
- Frontend: y-axis labels were clipping their leading digit (33 px of room for a ~63 px label); ticks now compact to '29k MW'. 'HOURS AHEAD' caption no longer collides with the tick row. Security headers added to vercel.json but NOT live.

Unfinished, in priority order

- 1. Risk panel is 7.0 days stale in production (read 2026-10-31 07:00Z beside a 1.9 h forecast). A weather-blind risk model is trained and tested but NOT wired into serving. This is the largest user-visible defect. Effort M, risk HIGH (it changes what users are told).
- 2. /docs, /redoc and /openapi.json are publicly exposed. Fix is a 3-argument FastAPI gate on an env var plus removing the footer link. Effort S, risk low.
- 3. /health mixes whole-history counters with served-frame currency, so a reader cannot tell which object a number describes. A diff exists, unapplied. Effort S.
- 4. tests/test_risk_snapshot.py:153 is wall-clock flaky (3.0 h lookalike threshold vs a 5-day-old local parquet). Inject a fixed clock; do not weaken the assertion. Effort S.
- 5. The retrain chain once exited 1 (models.train_baselines, 15:34:06Z) and the cause was never established. Effort S to diagnose, unknown to fix.
- 6. Accuracy/product backlog: IT_SOUTH is ~9.6% MAPE at h+24 against ES ~3.3% because the pooled model has no zone-identity feature; no rung of the utilisation ladder reaches the 0.6 alarm line (max 0.490872), so the extreme-overload alarm never fires; advance warning for marginal overloads, middle-band recalibration, coverage drift, an NWP weather input, rated capacity instead of a p99-of-load proxy, per-zone UI, a replay picker, and real Pakistani data are all still open. Each is M-L except the UI items (M).
- 7. No CI job runs the test suite or the integrity verifier on push. Effort S once check 2 above is resolved.

Risky, or needs your decision

- The unverified frontend chart fix: I would not merge it into main without one browser pass at 393x852, because main auto-deploys.
- The CSP in vercel.json is untested. A too-tight connect-src blanks the app; verify on a preview deployment before promoting.
- Wiring the weather-blind risk model changes the semantics of every risk number users see. It needs your call, and it should ship with the 'which artifact scored this' field.
- The committed feature parquet is now about 5 days behind the live frame. That is by design, but any doc sentence quoting one number for 'the data' is wrong until it names the object.
- The live API is request-driven: freshness depends on traffic. It advanced correctly during automated verification, but there is no scheduler, so a quiet hour serves an older origin.

Rules the next agent must keep

- Never change a gate threshold, metric, label rule or split to improve a number.
- Every figure in docs, UI or a resume point comes from a committed artifact or a command that was just run, and must name its object.
- Stop and ask before retraining; never call train_and_evaluate() for measurement.
- No push to main without review; no deploy without approval; no invented claims such as 'live' or 'Pakistan-ready'.